

Project Title-
CUSTOMER PURCHASE BEHAVIOR ANALYSIS USING EXPLORATORY DATA ANALYSIS (EDA) AND MICROSOFT POWER BI

Submitted in partial fulfillment of the requirements for the award of the degree of

Master of Computer Application (MCA)

By

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Executive Summary

Customer behavior analysis has become a crucial component of modern business intelligence, enabling organizations to understand purchasing trends and make informed decisions. This project presents a comprehensive analysis of customer purchasing behavior using the Super Mart Sales dataset through Exploratory Data Analysis (EDA) and Microsoft Power BI.

The project involves data collection, preprocessing, data cleaning, feature engineering, and exploratory data analysis to identify meaningful patterns within retail sales data. Interactive dashboards were developed using Microsoft Power BI to visualize key business metrics such as sales, profit, customer segments, payment methods, shipping modes, product categories, and regional performance.

The findings indicate that customer purchasing behavior varies significantly across customer segments, geographical regions, product categories, and seasonal periods. Consumer customers contributed the highest sales, while the West region demonstrated the strongest business performance. Office Supplies emerged as the highest-selling product category, and Standard Class shipping remained the most preferred delivery method.

The developed dashboard provides decision-makers with a powerful analytical platform for monitoring business performance, identifying opportunities, and supporting strategic planning through data-driven insights.

Keywords: Customer Behavior Analysis, Business Intelligence, Exploratory Data Analysis, Microsoft Power BI, Dashboard, Data Visualization, Retail Analytics.

- **Topic of the study-**
  The project titled **"Customer Purchase Behavior Analysis Using Exploratory Data Analysis (EDA) and Microsoft Power BI"** focuses on analyzing retail sales data to understand customer purchasing behavior, identify business trends, and generate meaningful insights using interactive dashboards and data visualization techniques.

- **Objective-**
  - To analyze customer purchasing behavior using the Super Mart Sales dataset.
  - To perform data cleaning and Exploratory Data Analysis (EDA).
  - To identify sales trends, customer segments, payment preferences, and product performance.
  - To develop an interactive dashboard using Microsoft Power BI.
  - To provide business insights that support data-driven decision-making.

- **System Analysis-**
  The existing system relies on manual reporting and static spreadsheets, making it difficult to analyze customer behavior efficiently. The proposed system uses Microsoft Power BI to automate data analysis, visualize business performance, and provide interactive reports for better decision-making.

- **System Design-**
  The system was designed by collecting the retail dataset, preprocessing the data, performing data cleaning, creating relationships, applying DAX measures, and designing interactive dashboard visualizations such as KPI cards, line charts, bar charts, donut charts, and maps.

- **System Implementation-**
  The implementation involved importing the dataset into Microsoft Power BI, cleaning the data using Power Query, creating calculated measures with DAX, and developing an interactive dashboard that displays sales trends, profit analysis, regional performance, customer segments, shipping modes, payment methods, and product categories.

- **Testing-**
  The developed dashboard was tested to ensure data accuracy, proper functioning of filters and slicers, correct calculation of KPIs, and consistency of visualizations. The results confirmed that the dashboard provides accurate and reliable business insights.

- **Future Scope**-
  - Integrate machine learning models for sales forecasting.
  - Develop customer recommendation and churn prediction systems.
  - Connect the dashboard with live cloud databases for real-time analysis.
  - Implement predictive analytics for business planning.
  - Enhance the dashboard with advanced AI-driven visualizations and automation.

Table of Contents

1. **Certificate**
2. **Declaration**
3. **Acknowledgement**
4. **Executive summary**
5. **Abstract**
6. **Chapter 1 – Introduction**
7. **Chapter 2 – System Study**
8. **Chapter 3 – System Analysis**
9. **Chapter 4 – System Design**
10. **Chapter 5 – System Implementation**
11. **Chapter 6 – Testing**
12. **Chapter 7 – Results and Discussion**
13. **Chapter 8 – Conclusion & Future Scope**
14. **Chapter 9 – References**
15. **Chapter 10 – Appendices**

Abstract

The present study entitled **"Customer Purchase Behavior Analysis Using Exploratory Data Analysis (EDA) and Microsoft Power BI"** focuses on understanding customer purchasing patterns and retail business performance using data analytics and visualization techniques. The project utilizes the **Super Mart Sales** dataset to identify trends in sales, profit, customer segments, product categories, payment methods, shipping modes, and regional performance. The primary objective is to transform raw transactional data into meaningful business insights that support effective decision-making.

The system analysis phase involved studying the existing retail sales reporting process and identifying its limitations, such as manual reporting, static analysis, and limited visualization capabilities. Based on these observations, a Business Intelligence solution was proposed using Microsoft Power BI to provide interactive dashboards and real-time analytical reports.

The system was designed by collecting and preprocessing the dataset, performing data cleaning, removing inconsistencies, and applying feature engineering techniques to improve data quality. Exploratory Data Analysis (EDA) was carried out using descriptive statistics and various visualization techniques to identify customer behavior, seasonal trends, sales distribution, and product performance.

The implementation phase involved developing an interactive Power BI dashboard containing Key Performance Indicators (KPIs), regional sales analysis, monthly sales trends, profit analysis, customer segmentation, payment mode analysis, shipping mode analysis, product category analysis, and geographical sales visualization. These visualizations provide an intuitive platform for monitoring organizational performance and customer purchasing behavior.

The developed system was tested by validating dashboard calculations, data accuracy, visualization performance, filter functionality, and report responsiveness. The dashboard successfully generated accurate analytical insights that can assist business managers in making informed strategic decisions.

The project concludes that Business Intelligence tools such as Microsoft Power BI significantly improve the process of analyzing retail sales data by providing interactive, accurate, and easy-to-understand visual reports. The findings indicate that customer purchasing behavior varies across regions, customer segments, product categories, and time periods, enabling organizations to optimize marketing strategies, inventory planning, and sales performance.

In the future, the system can be enhanced by integrating machine learning algorithms for sales forecasting, customer churn prediction, recommendation systems, and real-time data streaming. The dashboard may also relate to cloud-based databases to support large-scale business analytics and enterprise-level decision-making.

Chapter 1 : Introduction

1.1 Introduction

In today's digital era, organizations generate massive amounts of data through daily business operations. Every customer transaction, product purchase, payment, and shipment contribute to a continuously growing dataset. However, raw data alone has limited value unless it is transformed into meaningful information that supports business decisions. This transformation is achieved through Business Intelligence (BI) and data analytics techniques.

Customer behavior analysis is one of the most important applications of data analytics. It helps organizations understand how customers purchase products, identify their preferences, analyze buying patterns, and predict future trends. These insights enable businesses to improve customer satisfaction, optimize inventory, enhance marketing strategies, and increase profitability.

Microsoft Power BI is one of the most widely used Business Intelligence tools for creating interactive dashboards and reports. It enables users to transform raw data into visual representations such as charts, maps, graphs, and KPI cards, making complex datasets easier to understand.

This project, titled **"Customer Purchase Behavior Analysis Using Exploratory Data Analysis (EDA) and Microsoft Power BI,"** focuses on analyzing retail sales data from the **Super Mart Sales dataset**. The project applies data preprocessing, exploratory data analysis, and visualization techniques to study customer purchasing behavior and generate actionable business insights.

1.2 Background of the Study

The rapid growth of e-commerce and retail industries has increased the importance of customer-centric business strategies. Companies now rely on analytical tools to understand customer purchasing habits, identify profitable products, and optimize operational performance.

Traditionally, organizations depended on static reports and spreadsheets, which required significant manual effort and often delayed decision-making. Modern Business Intelligence platforms such as Microsoft Power BI overcome these limitations by providing dynamic dashboards that allow users to interact with data in real time.

The Super Mart Sales dataset provides an excellent opportunity to demonstrate how Business Intelligence can be applied to analyze customer behavior. The dataset contains information related to customer demographics, sales, profits, product categories, regions, shipping methods, and payment modes, making it suitable for comprehensive retail analysis.

1.3 Problem Statement

Retail organizations collect vast amounts of transactional data every day. However, much of this data remains underutilized due to the lack of efficient analytical tools. Traditional reporting methods make it difficult to identify customer purchasing patterns, evaluate product performance, monitor regional sales, and assess profitability.

The absence of interactive visualization tools often results in delayed decision-making and missed business opportunities. Therefore, there is a need for an analytical system that can transform raw sales data into meaningful visual insights.

This project addresses this problem by developing an interactive Power BI dashboard that enables organizations to analyze customer behavior effectively through Exploratory Data Analysis and Business Intelligence techniques.

1.4 Objectives of the Study

The primary objective of this project is to analyze customer purchasing behavior using Exploratory Data Analysis and Microsoft Power BI.

Specific Objectives

- To collect and preprocess retail sales data.

- To perform Exploratory Data Analysis (EDA) on the dataset.
- To identify customer purchasing patterns.
- To analyze sales and profit trends.
- To examine regional sales performance.
- To study customer segmentation.
- To evaluate product category performance.
- To analyze payment and shipping preferences.
- To develop an interactive Power BI dashboard.
- To generate business recommendations based on analytical findings.

1.5 Scope of the Study

The scope of this project includes the analysis of historical retail sales data using Business Intelligence tools. The study focuses on identifying customer purchasing behaviour, sales trends, and business performance through data visualization.

The project is limited to descriptive analytics and does not include predictive modelling or machine learning algorithms. However, it provides a strong foundation for future analytical enhancements.

The developed dashboard can assist business managers, analysts, and decision-makers in monitoring organizational performance and identifying opportunities for business growth.

1.6 Need for the Study

Understanding customer behavior has become essential for organizations operating in competitive markets. Businesses must identify customer preferences, monitor purchasing trends, and evaluate product performance to remain competitive.

This study is important because it:

- Improves business decision-making.

- Supports customer-centric marketing strategies.
- Enhances inventory management.
- Identifies profitable products.
- Monitors regional performance.
- Reduces manual reporting efforts.
- Provides interactive business insights.

1.7 Significance of the Study

The significance of this project lies in its practical application of Business Intelligence techniques to real-world retail data. The interactive dashboard enables users to monitor sales performance, profitability, customer behavior, and product trends efficiently.

The project demonstrates how Power BI can transform complex datasets into meaningful visualizations, enabling organizations to make faster and more informed decisions.

The findings of this study may also serve as a reference for students, researchers, and organizations interested in retail analytics and customer behavior analysis.

1.8 Existing System

Many organizations still rely on traditional reporting methods using spreadsheets and static reports. These approaches require manual data processing and often fail to provide real-time insights.

Limitations of the Existing System

- Static reports with limited interactivity.
- Time-consuming data analysis.
- Difficulty identifying customer behavior patterns.

- Limited visualization capabilities.
- Delayed decision-making.
- High dependency on manual reporting.

1.9 Proposed System

The proposed system utilizes Microsoft Power BI to create an interactive dashboard for customer purchase behavior analysis.

The system integrates:

- Data collection.
- Data cleaning.
- Exploratory Data Analysis.
- Dashboard development.
- Interactive visualization.
- Business insight generation.

Advantages of the Proposed System

- Interactive dashboards.
- Dynamic filtering.
- Real-time analysis.
- Better visualization.
- Improved decision support.
- Easy report generation.

1.10 Research Methodology

The project follows a systematic methodology to ensure accurate analysis and meaningful visualization of customer purchase behavior. Each phase of the methodology contributes towards achieving the project objectives effectively.

Step 1: Data Collection

The Super Mart Sales dataset was selected as the primary data source for this project. The dataset contains detailed information regarding customer transactions, sales amount, profit, quantity sold, shipping methods, product categories, customer segments, payment modes, and geographical regions. The dataset was imported into Microsoft Power BI for further analysis.

Step 2: Data Cleaning

Data preprocessing is an essential step before analysis. The collected dataset was cleaned using **Power Query Editor** in Microsoft Power BI. During this process:

- Duplicate records were identified and removed.
- Missing or incomplete values were handled appropriately.
- Incorrect data formats were corrected.
- Column names were standardized.
- Unnecessary columns were removed to improve processing efficiency.

These preprocessing activities improved data quality and ensured accurate analytical results.

Step 3: Data Transformation

After cleaning, the dataset was transformed according to the project requirements. Additional calculated columns and measures were created using **Data Analysis Expressions (DAX)** to support business analysis.

The transformation process included:

- Creating calculated measures for Average Sales and Average Profit.

- Calculating Average Quantity Sold.
- Computing Average Delivery Days.
- Organizing product categories and customer segments.
- Preparing date fields for monthly and yearly trend analysis.

These transformations enhanced the analytical capability of the dashboard.

Step 4: Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the characteristics of the dataset and identify meaningful patterns.

The following analyses were conducted:

- Sales trend analysis
- Profit trend analysis
- Customer segment analysis
- Regional sales analysis
- Product category analysis
- Sub-category performance analysis
- Shipping mode analysis
- Payment mode analysis
- State-wise sales analysis

Various charts and statistical techniques were used to summarize and interpret the data effectively.

Step 5: Dashboard Development

After completing the EDA process, an interactive dashboard was developed using Microsoft Power BI.

The dashboard consists of:

- KPI Cards
- Line Charts
- Bar Charts
- Donut Charts
- Map Visualization
- Interactive Region Filters (Slicers)

The dashboard enables users to explore business performance dynamically through interactive filters and visual representations.

Step 6: Business Insight Generation

The visualizations generated through Power BI were carefully analyzed to identify significant business insights. Customer purchasing behavior, regional performance, product demand, and seasonal sales trends were evaluated to provide meaningful recommendations for decision-makers.

Step 7: Documentation

Finally, all findings, visualizations, observations, and recommendations were documented systematically in this project report. The report includes methodology, analysis, dashboard interpretation, conclusions, and future scope.

1.11 Tools and Technologies Used

The successful implementation of this project required the use of various software tools and technologies. Each tool played an important role in different stages of the project.

Software/Tool	**Purpose**
Microsoft Power BI Desktop	Dashboard Development and Visualization

Microsoft Excel	Dataset Storage and Initial Data Preparation
Power Query Editor	Data Cleaning and Transformation
DAX (Data Analysis Expressions)	Creating KPIs and Calculated Measures
Microsoft Word	Report Documentation
Windows 11	Operating System

1.12 Hardware Requirements

The project was developed using the following hardware configuration.

Component	Specification
Laptop	MSI Cyborg 15 A12UCX
Processor	Intel Core i5-12450H (12th Generation)
RAM	8 GB DDR5
Graphics	NVIDIA GeForce RTX 2050 (4 GB)
Storage	477 GB SSD
Operating System	Windows 11 (64-bit)

1.13 Software Requirements

The following software tools were used during the development of the project.

Software	Version
Microsoft Power BI Desktop	Latest Stable Version
Microsoft Excel	Microsoft Office 365
Microsoft Word	Microsoft Office 365
Windows Operating System	Windows 11
Power Query	Built-in with Power BI
DAX Language	Built-in with Power BI

1.14 Organization of the Report

The project report is organized into ten chapters, each describing a different phase of the development of the **Super Mart Sales Dashboard using Microsoft Power BI**.

- **Chapter 1 – Introduction:** Presents the background of the study, problem statement, objectives, scope, significance, research methodology, and organization of the report.
- **Chapter 2 – System Study:** Describes the existing system, identifies its limitations, explains the proposed system, presents the feasibility study, and reviews the relevant literature.
- **Chapter 3 – System Analysis:** Discusses the system requirements, functional and non-functional requirements, data flow diagrams, use case diagrams, flowcharts, and overall system analysis.
- **Chapter 4 – System Design:** Explains the system architecture, input and output design, data model, dashboard layout, Power Query transformations, DAX measures, user interface, and navigation design.
- **Chapter 5 – System Implementation:** Describes the implementation process, including data import, data cleaning, transformation, data modeling, DAX calculations, dashboard creation, and implementation of interactive visualizations using Microsoft Power BI.
- **Chapter 6 – Testing:** Presents the testing methodology, test cases, validation process, performance testing, user acceptance testing, and testing results to verify the functionality and reliability of the dashboard.
- **Chapter 7 – Results and Discussion:** Analyzes the dashboard outputs, interprets the visualizations, discusses key findings, and presents business insights and recommendations based on the sales data.
- **Chapter 8 – Conclusion and Future Scope:** Summarizes the project, highlights its achievements, discusses limitations, and suggests future enhancements such as real-time data integration, predictive analytics, and AI-based forecasting.

- **Chapter 9 – References:** Lists the books, research papers, journals, official Microsoft documentation, and online resources referred to during the development of the project.
- **Chapter 10 – Appendices:** Contains supporting materials such as dashboard screenshots, dataset sample, data model view, DAX measures, test case summary, user manual, dashboard navigation guide, and project source files.

1.15 Chapter Summary

This chapter introduced the project by explaining the importance of customer purchase behavior analysis in the retail industry. It discussed the project background, problem statement, objectives, scope, significance, existing and proposed systems, research methodology, hardware and software requirements, and report organization. The chapter establishes the foundation for the remaining chapters by outlining the overall structure and purpose of the project.

Chapter 2: System Study

2.1 Introduction

A system study is one of the most important phases of system development as it provides a comprehensive understanding of the existing business process, identifies the limitations of the current system, and establishes the requirements for the proposed system. Before designing any software solution, it is essential to analyze how the existing system functions, what problems users encounter, and how these problems can be resolved through modern technologies.

In today's competitive business environment, organizations generate enormous volumes of data from customer transactions, inventory management, product sales, shipping activities, and payment systems. Every purchase made by a customer contributes valuable information that can help organizations understand customer preferences, market trends, product demand, and business performance. However, simply collecting data is not sufficient. The data must be organized, processed, analyzed, and visualized to extract meaningful business insights.

Traditionally, retail organizations have relied on spreadsheets, manual reports, and static graphs to monitor business performance. Although these methods provide basic reporting capabilities, they fail to support dynamic analysis and interactive exploration of business data. Managers often spend considerable time preparing reports, filtering data manually, and identifying sales patterns. These traditional approaches are prone to human error, consume significant time, and limit the organization's ability to make timely business decisions.

Business Intelligence (BI) has transformed the way organizations analyze business data. Microsoft Power BI is one of the most widely used Business Intelligence platforms that enables users to convert raw data into interactive dashboards and meaningful visualizations. Through features such as drill-down analysis, dynamic filtering, geographical mapping, and real-time reporting, Power BI simplifies the decision-making process.

The present project utilizes Microsoft Power BI to analyze customer purchase behavior using the Super Mart Sales dataset. The dashboard developed in this project enables users

to monitor sales trends, customer segments, product categories, payment modes, shipping methods, and regional performance through interactive visualizations. The system also applies Exploratory Data Analysis (EDA) techniques to identify hidden patterns and generate actionable business insights.

The system study conducted for this project focuses on understanding the current retail sales reporting process, evaluating its shortcomings, and proposing a Business Intelligence-based solution capable of improving data analysis, reporting efficiency, and decision-making.

2.2 Study of Existing System

The existing system used in many retail organizations is primarily based on manual data processing and spreadsheet applications such as Microsoft Excel. Sales data generated from customer transactions is stored in tabular format, and business reports are prepared manually by filtering and summarizing the data.

Managers typically generate monthly, quarterly, and annual reports to evaluate sales performance. However, the preparation of these reports requires significant manual effort. Data from multiple sources must be consolidated, cleaned, and analyzed before meaningful reports can be generated. As the volume of data increases, managing and analyzing the information becomes increasingly difficult.

The traditional reporting system lacks interactive visualization features. Users are unable to drill down into specific regions, customer segments, or product categories. Consequently, identifying customer purchasing behavior and understanding business performance becomes a challenging task.

The existing system generally performs the following activities:

- Collection of customer transaction data.
- Storage of sales information in spreadsheets.
- Manual calculation of sales and profit.
- Preparation of summary reports.

- Limited graphical representation using static charts.
- Manual comparison of monthly and yearly sales.
- Preparation of reports for management review.

Although the existing system provides basic reporting functionality, it does not support real-time business intelligence or interactive analytics.

2.2.1 Working of the Existing System

The workflow of the existing retail reporting system is illustrated below:

1. Customer purchases products from the retail store.
2. Sales transactions are recorded in the database or spreadsheet.
3. Data is exported to Microsoft Excel.
4. Reports are prepared manually using formulas and pivot tables.
5. Charts are generated for management review.
6. Business decisions are made based on static reports.

This process requires repeated manual intervention whenever new sales data becomes available.

2.2.2 Drawbacks of the Existing System

Despite serving as the primary reporting mechanism, the existing system suffers from several limitations that reduce business efficiency.

1. Manual Report Generation

The preparation of reports requires manual filtering, calculations, and formatting, increasing the possibility of human error.

2. Time-Consuming Analysis

Business managers spend considerable time preparing reports instead of analyzing business performance.

3. Lack of Interactive Visualization

The existing system provides only static charts that cannot be filtered dynamically.

4. Poor Customer Behavior Analysis

Customer purchasing patterns cannot be analyzed efficiently because the reports lack segmentation and interactive exploration.

5. Limited Scalability

As business data grows, spreadsheets become increasingly difficult to manage and process.

6. Difficulty in Identifying Business Trends

Seasonal trends, regional performance, and product demand are difficult to identify through manual reports.

7. Delayed Decision Making

Since reports are generated periodically rather than in real time, business decisions are often delayed.

8. High Dependency on Human Resources

The organization depends heavily on skilled personnel to prepare reports manually.

2.2.3 Problems Identified in the Existing System

The detailed system study identified several practical problems in the traditional reporting process.

Problem	Impact on Organization
Manual reporting	Increased workload

Static reports	Limited business insights
Large datasets	Slow processing
Spreadsheet dependency	Higher chances of errors
No automation	Delayed reporting
Poor visualization	Difficult trend identification
Limited filtering	Reduced analytical capability
Lack of dashboards	Inefficient decision-making

2.3 Need for the Proposed System

The limitations identified in the existing system highlight the need for a modern Business Intelligence solution capable of transforming raw business data into meaningful visual information.

Organizations require a system that not only stores data but also analyzes it efficiently and presents actionable insights to decision-makers. The proposed Power BI dashboard fulfills this requirement by integrating data preprocessing, Exploratory Data Analysis (EDA), interactive visualizations, and business intelligence techniques into a single analytical platform.

The proposed system enables users to monitor business performance in real time, compare regional sales, analyze customer purchasing behavior, evaluate product categories, and identify business opportunities through dynamic reports.

Unlike the existing system, the proposed dashboard provides interactive filtering, drill-down analysis, geographical mapping, KPI monitoring, and automated report generation. These features significantly reduce manual effort and improve the overall efficiency of business analysis.

2.4 Proposed System

The proposed system is designed to overcome the shortcomings of the existing manual reporting system by utilizing **Microsoft Power BI** as the primary Business Intelligence tool. The system focuses on analyzing customer purchase behavior through interactive dashboards and Exploratory Data Analysis (EDA). Instead of relying on static reports, the proposed solution enables dynamic data visualization, allowing users to interact with the data through filters, slicers, and graphical reports.

The system begins by importing the Super Mart Sales dataset into Microsoft Power BI. The dataset is then processed using **Power Query Editor**, where unnecessary columns are removed, data types are corrected, duplicate records are eliminated, and missing values are handled. This preprocessing stage ensures that the dataset is clean, consistent, and suitable for analysis.

After preprocessing, relationships among different data fields are established, and **DAX (Data Analysis Expressions)** measures are created to calculate important Key Performance Indicators (KPIs) such as Average Sales, Average Profit, Average Quantity Sold, and Average Delivery Days. These measures provide summarized information that helps users evaluate business performance quickly.

The processed data is then visualized using multiple Power BI visuals, including bar charts, line charts, donut charts, KPI cards, and geographical maps. The dashboard allows users to analyze sales trends across different months, compare profits, evaluate customer segments, examine payment modes, identify the most profitable product categories, and monitor sales performance across different states and regions.

One of the major advantages of the proposed system is its interactivity. Users can select a specific region using the slicer available on the dashboard, and all visualizations update automatically based on the selected region. This feature enables managers to perform detailed regional analysis without generating separate reports.

The proposed system significantly reduces manual effort, improves reporting accuracy, enhances visualization capabilities, and supports data-driven decision-making. It provides an efficient platform for understanding customer purchasing behavior and identifying opportunities for business growth.

2.4.1 Features of the Proposed System

The developed Power BI dashboard includes several advanced features that improve the efficiency of business analysis. These features enable organizations to monitor business performance through interactive visualizations and summarized analytical reports.

Interactive Dashboard

The dashboard provides a single interface where users can monitor various business metrics simultaneously. All visualizations are interconnected, allowing dynamic exploration of sales data.

KPI Cards

Key Performance Indicators (KPIs) display important business metrics, including:

- Average Sales
- Average Profit
- Average Quantity
- Average Delivery Days

These KPIs provide a quick overview of overall business performance.

Regional Analysis

The dashboard analyzes sales across different geographical regions such as East, West, South, and Central. This enables organizations to compare regional performance and identify high-performing markets.

Monthly Sales and Profit Analysis

Line charts are used to visualize monthly sales and monthly profit trends. These charts help identify seasonal demand, peak sales periods, and fluctuations in profitability.

Customer Segment Analysis

The dashboard classifies customers into different segments such as Consumer, Corporate, and Home Office. This helps organizations understand which customer group contributes the highest revenue.

Product Category Analysis

Sales are analyzed across different product categories including Furniture, Office Supplies, and Technology. The dashboard identifies the highest-selling product categories and their contribution to total sales.

Sub-Category Analysis

Detailed analysis of product sub-categories such as Phones, Chairs, Tables, and Storage enables businesses to identify their most profitable products.

Payment Mode Analysis

The dashboard displays customer payment preferences using donut charts. Organizations can analyze whether customers prefer Cash on Delivery (COD), Online Payment, or Card Payment.

Shipping Mode Analysis

The dashboard also analyzes customer preferences regarding different shipping methods, such as Standard Class, Second Class, First Class, and Same Day Delivery.

Geographic Sales Visualization

The integrated map visualization displays state-wise sales performance, allowing users to identify regions generating the highest revenue.

2.4.2 Comparison between Existing and Proposed System

Existing System	Proposed System
Manual report generation	Automated dashboard reporting
Static Excel reports	Interactive Power BI dashboard

Time-consuming analysis	Faster real-time analysis
Limited visualization	Advanced interactive visualizations
Manual calculations	Automated KPI calculations
Difficult to identify trends	Easy trend identification
No interactive filtering	Dynamic filters and slicers
Limited business insights	Comprehensive business intelligence
High chances of errors	Improved accuracy
Delayed decision-making	Faster data-driven decisions

Table 2.1: Comparison between Existing and Proposed System

2.5 Objectives of the Proposed System

The proposed system has been developed with the objective of providing an efficient Business Intelligence solution for analyzing customer purchase behavior. The major objectives are:

- To automate the process of business reporting.
- To reduce manual effort in sales analysis.
- To perform Exploratory Data Analysis on retail sales data.
- To identify customer purchasing patterns.
- To monitor regional sales performance.
- To analyze customer segments and product categories.
- To evaluate payment and shipping preferences.
- To create interactive dashboards using Microsoft Power BI.
- To support business managers in making informed decisions.

- To improve organizational efficiency through data visualization.

2.6 Feasibility Study

Before implementing the proposed system, a feasibility study was conducted to evaluate whether the project could be developed successfully with the available resources. The study ensures that the system is technically, economically, operationally, legally, and practically feasible.

2.6.1 Technical Feasibility

The proposed system is technically feasible because Microsoft Power BI provides all the necessary tools required for data cleaning, transformation, visualization, and dashboard development. The software supports integration with Microsoft Excel and offers built-in capabilities such as Power Query, DAX, and interactive visualizations.

The project was developed on a laptop equipped with an Intel Core i5 processor, 8 GB RAM, SSD storage, and Windows 11 operating system, which satisfies the hardware requirements of Microsoft Power BI Desktop. Therefore, no additional hardware investment was required for implementation.

2.6.2 Economic Feasibility

The project is economically feasible because Microsoft Power BI Desktop is available free of cost for educational use. The Super Mart Sales dataset is publicly available, eliminating any data acquisition costs. Existing hardware resources were sufficient, and no specialized software licenses or additional infrastructure were required. As a result, the overall development cost remained minimal.

2.6.3 Operational Feasibility

The proposed system is operationally feasible as it is designed with a simple and user-friendly interface. Users with basic computer knowledge can easily operate the dashboard using filters, slicers, and interactive visuals. The dashboard reduces manual effort and improves reporting efficiency, making it suitable for day-to-day business analysis.

Chapter 3 : System Analysis

3.1 Introduction

System Analysis is a systematic process of examining an existing system, understanding its operational procedures, identifying its limitations, and determining the requirements for developing an improved solution. It serves as one of the most critical phases in the System Development Life Cycle (SDLC) because it establishes the foundation upon which the entire system is designed and implemented.

In the context of this project, **"Customer Purchase Behavior Analysis using Exploratory Data Analysis (EDA) and Microsoft Power BI,"** system analysis focuses on understanding the retail sales reporting process, identifying analytical requirements, and determining how Business Intelligence can improve business decision-making. The analysis phase involves studying the available sales dataset, understanding customer purchasing patterns, identifying the information needs of business managers, and defining the functional and non-functional requirements of the proposed dashboard.

Retail organizations generate a significant amount of transactional data through daily sales activities. These transactions include customer information, purchased products, order quantities, payment methods, shipment details, regional sales, and profits. Although the data is valuable, organizations often fail to extract meaningful insights due to reliance on traditional reporting techniques. Manual analysis using spreadsheets becomes increasingly difficult as the volume of data grows.

The objective of system analysis is to identify the shortcomings of these traditional methods and propose an analytical framework capable of transforming raw data into meaningful business intelligence. Through this phase, user requirements, business requirements, and system constraints are carefully studied before designing the dashboard.

The analysis performed in this project ensures that the proposed system addresses real business problems such as delayed reporting, poor visualization, difficulty in identifying customer trends, and inefficient decision-making. By clearly defining system requirements during this phase, the development of the Power BI dashboard becomes more organized, efficient, and aligned with organizational objectives.

3.2 Requirement Analysis

Requirement analysis is the process of identifying the needs and expectations of stakeholders before the development of the proposed system. It ensures that the final dashboard fulfills all business objectives and provides the analytical capabilities required by users. Proper requirement analysis minimizes development errors, improves system quality, and ensures that the final solution addresses practical business problems.

For this project, the requirements were identified by examining the Super Mart Sales dataset, studying retail reporting practices, and understanding the objectives of customer purchase behavior analysis. The requirements were classified into business requirements, user requirements, functional requirements, non-functional requirements, and data requirements.

The analysis considered the needs of managers who require quick access to business information, sales executives who monitor sales performance, and analysts who examine customer purchasing patterns. The dashboard was therefore planned to provide interactive reports, visual summaries, and meaningful insights instead of static reports.

The requirement analysis also focused on identifying the key performance indicators (KPIs) required for evaluating business performance. Metrics such as average sales, average profit, average quantity sold, regional sales distribution, category-wise sales, customer segmentation, and payment mode analysis were selected because they directly support business decision-making.

3.3 User Requirements

The proposed dashboard has been designed to satisfy the requirements of different categories of users involved in retail business management. Each user interacts with the dashboard according to their responsibilities and information needs.

Business Manager

The Business Manager requires summarized reports to evaluate overall organizational performance. The manager should be able to monitor sales, profits, customer segments, and regional performance without manually preparing reports.

The dashboard should allow the manager to:

- Monitor overall sales performance.
- Compare profits across different regions.
- Identify top-performing product categories.
- Analyze monthly sales trends.
- Support strategic business planning.

Sales Manager

The Sales Manager focuses on monitoring sales activities and identifying opportunities for improving revenue generation.

The dashboard should enable the Sales Manager to:

- Compare regional sales.
- Monitor monthly sales performance.
- Analyze product demand.
- Identify profitable products.
- Track customer purchasing behavior.

Business Analyst

The Business Analyst requires detailed data exploration to identify hidden trends and business patterns.

The dashboard should support:

- Exploratory Data Analysis.
- Customer segmentation.

- Product category comparison.
- Payment mode analysis.
- Shipping mode analysis.
- Trend analysis using interactive filters.

3.4 Business Requirements

Business requirements describe the organizational objectives that the proposed system is expected to achieve. These requirements are directly related to improving business performance through data-driven decision-making.

The major business requirements include:

- Improve business reporting efficiency.
- Reduce dependency on manual spreadsheet reports.
- Provide interactive business dashboards.
- Identify customer purchasing behavior.
- Analyze sales and profit performance.
- Monitor state-wise and regional sales.
- Evaluate payment and shipping preferences.
- Improve inventory planning through category analysis.
- Support management in making strategic decisions.
- Generate actionable business insights.

The fulfillment of these requirements enables organizations to improve operational efficiency, customer satisfaction, and overall profitability.

3.5 Data Requirements

The effectiveness of the dashboard depends on the availability of accurate and relevant data. Therefore, analyzing the dataset requirements is an important aspect of system analysis.

The Super Mart Sales dataset contains multiple attributes that are essential for customer purchase behavior analysis. These attributes include customer details, product information, sales values, profits, quantities, payment methods, shipping details, and regional information.

The major data elements required for the project are listed below.

Data Field	**Purpose**
Order ID	Unique identification of orders
Order Date	Monthly trend analysis
Ship Date	Delivery analysis
Customer Name	Customer identification
Segment	Customer segmentation
State	Regional analysis
Category	Category-wise sales
Sub-Category	Product analysis
Sales	Revenue calculation
Profit	Profitability analysis
Quantity	Purchase quantity analysis
Payment Mode	Customer payment preference
Ship Mode	Shipping preference analysis

The quality of these data fields directly influences the accuracy of the dashboard, and the business insights generated from it.

3.6 Functional Requirements

Functional requirements describe the specific operations and services that the proposed system must perform to meet user expectations. These requirements define the core functionality of the Power BI dashboard and ensure that it provides meaningful analysis of customer purchasing behavior.

The dashboard developed in this project has been designed to perform several analytical functions that support business intelligence and decision-making. The functional requirements are explained below.

3.6.1 Data Import

The system should allow the user to import the Super Mart Sales dataset from Microsoft Excel into Microsoft Power BI Desktop. The import process should preserve the original data structure and ensure that all records are available for analysis.

3.6.2 Data Cleaning

The system should provide facilities for cleaning the dataset using Power Query Editor. This includes removing duplicate records, correcting inconsistent data types, handling missing values, renaming columns where necessary, and ensuring data quality before analysis.

3.6.3 Data Transformation

The imported dataset should be transformed into a suitable format for analysis. The system should support calculated columns, derived fields, and DAX measures to generate meaningful business metrics.

3.6.4 Dashboard Generation

The system should generate an interactive dashboard consisting of KPI cards, bar charts, line charts, donut charts, maps, and slicers. These visualizations should present business information in a clear and understandable manner.

3.6.5 Sales Analysis

The dashboard should analyze monthly and yearly sales trends, helping users identify seasonal variations and growth patterns. Users should be able to compare sales across different months and years.

3.6.6 Profit Analysis

The system should calculate and visualize profit trends to identify profitable and less profitable periods. It should also support comparison of profits across product categories and regions.

3.6.7 Customer Segment Analysis

The dashboard should classify customers into Consumer, Corporate, and Home Office segments. Users should be able to compare the contribution of each segment toward overall sales.

3.6.8 Regional Analysis

The system should display regional sales distribution and allow users to compare performance across the East, West, South, and Central regions.

3.6.9 Product Analysis

The system should analyze product categories and sub-categories to identify high-performing and low-performing products. This helps management optimize inventory and marketing strategies.

3.6.10 Payment and Shipping Analysis

The dashboard should evaluate customer payment preferences and shipping methods. This information helps organizations improve customer service and logistics planning.

Table 3.1 Functional Requirements

Requirement ID	Functional Requirement	Description
FR-01	Import Dataset	Import Excel sales dataset into Power BI
FR-02	Clean Data	Remove duplicates and inconsistencies
FR-03	Transform Data	Create calculated columns and DAX measures
FR-04	Generate Dashboard	Develop interactive dashboard
FR-05	Analyze Sales	Monthly and yearly sales analysis
FR-06	Analyze Profit	Profit comparison and trend analysis
FR-07	Customer Segmentation	Analyze customer groups
FR-08	Product Analysis	Category and sub-category analysis
FR-09	Payment Analysis	Analyze payment modes
FR-10	Regional Analysis	Compare performance across regions

3.7 Non-Functional Requirements

Non-functional requirements specify the quality characteristics of the system. They describe how the system should perform rather than what functions it should provide.

Performance

The dashboard should process the available dataset efficiently and display reports with minimal loading time. Interactive filters and slicers should update all visualizations quickly without affecting system performance.

Reliability

The system should consistently provide accurate analytical results. KPI calculations, charts, and graphs must correctly reflect the underlying dataset whenever it is refreshed.

Usability

The dashboard should be simple, interactive, and easy to understand. Even users with limited technical knowledge should be able to navigate through the reports without difficulty.

Scalability

The system should be capable of handling larger datasets in the future. As the organization's sales records grow, the dashboard should continue to perform effectively without requiring significant redesign.

Security

The dataset should be protected against unauthorized modifications. If deployed in a real business environment, appropriate access controls should be implemented to maintain data confidentiality.

Maintainability

The dashboard should be easy to maintain and update. New data can be imported, KPIs can be modified, and additional visualizations can be incorporated with minimal effort.

Portability

The Power BI report should be easily transferable between different computers and should operate on supported Windows systems without requiring extensive configuration.

Table 3.2 Non-Functional Requirements

Requirement	Description
Performance	Fast loading and response time
Reliability	Accurate reporting and calculations
Usability	Easy-to-use interface

Scalability	Supports larger datasets
Security	Protects business data
Maintainability	Easy updates and modifications
Portability	Compatible with supported systems

3.8 Problem Analysis

The analysis of the existing reporting system revealed several challenges that affect the efficiency of business operations. Traditional reporting methods depend heavily on spreadsheets and manual calculations, making it difficult to obtain timely and accurate insights.

One of the major problems is the inability to visualize business data interactively. Static reports limit users to predefined summaries and do not allow dynamic exploration of sales trends or customer behavior. As the volume of retail data increases, manual analysis becomes increasingly complex and time-consuming.

The absence of automated reporting also delays managerial decision-making. Business managers often spend considerable time preparing reports instead of analyzing them. This affects the organization's ability to respond quickly to changing market conditions.

Another important challenge is identifying customer purchasing patterns. Traditional reports cannot effectively highlight customer segments, payment preferences, shipping behavior, or regional performance. As a result, organizations miss valuable opportunities to improve marketing strategies and customer satisfaction.

These problems justify the development of a Business Intelligence dashboard capable of providing interactive visualizations and meaningful analytical insights.

3.9 System Constraints

Although the proposed system offers several advantages, it also has certain limitations that should be considered during implementation.

- The dashboard is based on historical sales data and does not perform real-time analysis.
- The accuracy of reports depends entirely on the quality of the input dataset.
- Predictive analytics and machine learning models are beyond the scope of this project.
- Internet connectivity may be required for certain Power BI features if reports are published online.
- The project focuses only on descriptive analytics using Exploratory Data Analysis.

Despite these constraints, the developed dashboard effectively fulfills the objectives of customer purchase behavior analysis.

3.10 Expected Benefits of System Analysis

The system analysis phase provides several benefits that contribute to the successful development of the proposed dashboard.

- Clear understanding of business requirements.
- Identification of user expectations.
- Improved planning of dashboard features.
- Reduction in development errors.
- Better allocation of project resources.
- Enhanced communication between stakeholders.
- Strong foundation for system design and implementation.
- Increased probability of successful project completion.

3.11 Chapter Summary

This chapter presented a comprehensive analysis of the proposed Customer Purchase Behavior Analysis System. It examined the business requirements, user requirements, data requirements, functional requirements, and non-functional requirements necessary for developing an effective Business Intelligence solution. The chapter also analyzed the limitations of the existing reporting system, identified key business problems, discussed system constraints, and highlighted the expected benefits of the analysis phase.

Chapter 4: System Design

4.1 Introduction

System Design is one of the most important phases of the System Development Life Cycle (SDLC). After the completion of system analysis, the next step is to convert the identified requirements into a structured design that serves as the blueprint for system implementation. A well-designed system ensures that the final solution is efficient, reliable, user-friendly, and capable of meeting the business objectives.

In this project, **"Customer Purchase Behavior Analysis using Exploratory Data Analysis (EDA) and Microsoft Power BI,"** the system design phase focuses on designing an interactive Business Intelligence dashboard that transforms raw retail sales data into meaningful visual insights. The design process includes planning the overall architecture, defining the flow of data, selecting suitable visualizations, designing Key Performance Indicators (KPIs), creating dashboard layouts, and ensuring smooth interaction between different dashboard components.

The primary objective of the system design is to present business information in a way that enables users to understand customer purchasing behavior quickly and make informed decisions. Since retail organizations generate large volumes of transactional data, the dashboard must simplify complex datasets through effective visualization techniques.

The proposed design emphasizes simplicity, interactivity, and scalability. Users can filter data based on different regions and instantly observe changes across all visualizations. This interactive capability enhances analytical efficiency and supports faster business decision-making.

The dashboard has been designed using Microsoft Power BI because of its powerful visualization capabilities, seamless data integration, and support for advanced analytical features such as Power Query and DAX calculations. The final design provides managers with a comprehensive overview of sales performance, profitability, customer segments, shipping methods, payment preferences, and regional performance through a single interactive interface.

The following sections describe the overall system architecture, dashboard components, data flow, module design, visualization strategy, and interface design adopted during the development of the project.

4.2 Design Objectives

The design objectives define the goals that guided the development of the proposed dashboard. These objectives ensure that the final system is capable of meeting user requirements while maintaining high usability and performance.

The major design objectives are as follows:

- To develop an interactive Business Intelligence dashboard using Microsoft Power BI.
- To provide a centralized platform for customer purchase behavior analysis.
- To convert raw sales data into meaningful visualizations.
- To support quick and accurate business decision-making.
- To simplify complex retail datasets through graphical representation.
- To design a user-friendly interface with easy navigation.
- To enable dynamic filtering using slicers.
- To present sales, profit, quantity, and customer information through Key Performance Indicators (KPIs).
- To ensure that dashboard components update automatically based on user selections.
- To create a scalable dashboard that can accommodate future data updates.

These objectives guided the overall design process and ensured that the proposed system effectively fulfills the analytical requirements identified during the system analysis phase.

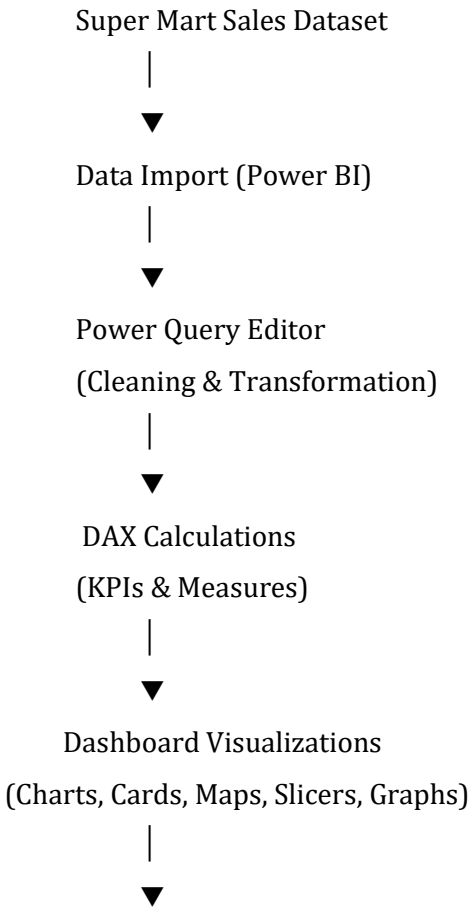
4.3 System Architecture

The overall architecture of the proposed system illustrates how data flows from the source dataset to the final interactive dashboard. It consists of several interconnected components that work together to transform raw retail data into meaningful business insights.

The architecture begins with the **Super Mart Sales dataset**, which serves as the primary source of information. The dataset is imported into Microsoft Power BI, where Power Query is used to clean and transform the data. After preprocessing, DAX measures are created to calculate business metrics such as average sales, average profit, and average quantity. The processed data is then used to generate interactive visualizations, including charts, maps, KPI cards, and slicers.

Finally, the dashboard presents these visualizations to the user, enabling dynamic analysis of customer purchasing behavior.

Figure 4.1 System Architecture



Business Insights

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Decision Making

Explanation

The architecture follows a sequential process beginning with data acquisition and ending with business decision-making. Each stage adds value by improving data quality, calculating meaningful metrics, and presenting the results through interactive visualizations. This layered architecture ensures that users receive accurate, reliable, and actionable business insights.

4.4 Data Flow Diagram (DFD)

A **Data Flow Diagram (DFD)** is a graphical representation that illustrates the movement of data within the system. It identifies the input source, processing activities, data storage, and output generated by the proposed system. The DFD helps in understanding how the retail sales data is transformed into meaningful business information through Microsoft Power BI.

In this project, the Super Mart Sales dataset serves as the primary input. The data undergoes several stages including data cleaning, transformation, Exploratory Data Analysis (EDA), and dashboard development before generating business insights for managers and decision-makers.

4.4.1 DFD Level 0 (Context Diagram)

The Level 0 DFD presents the system as a single process interacting with external entities.

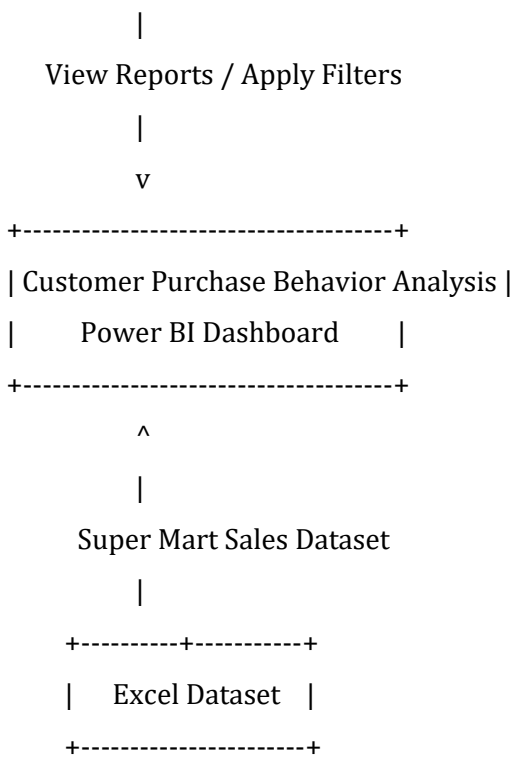
Figure 4.2: DFD Level 0

+----------------------+

| User |

| (Business Manager) |

+----------+-----------+



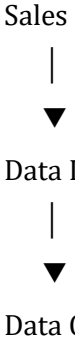
Explanation

The user interacts with the Power BI dashboard to explore reports and apply filters. The dashboard receives input from the Super Mart Sales dataset and processes the data to generate visualizations and business insights.

4.4.2 DFD Level 1

The Level 1 DFD provides a detailed representation of the internal processing stages.

Figure 4.3: DFD Level 1



(Power Query)

Data Transformation

DAX Calculations

EDA & Visualization

Interactive Dashboard



Business Insights

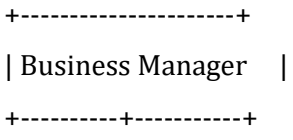
Explanation

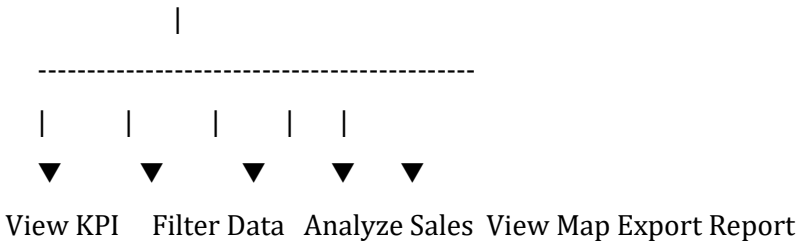
The sales dataset is first imported into Power BI. After cleaning and transformation using Power Query, DAX measures are created to calculate business KPIs. Exploratory Data Analysis is then performed to identify patterns and trends. Finally, interactive dashboards are generated to support business decision-making.

4.5 Use Case Diagram

A Use Case Diagram illustrates the interaction between the user and the proposed system. In this project, the Business Manager is the primary actor who interacts with the dashboard to perform various analytical tasks.

Figure 4.4: Use Case Diagram





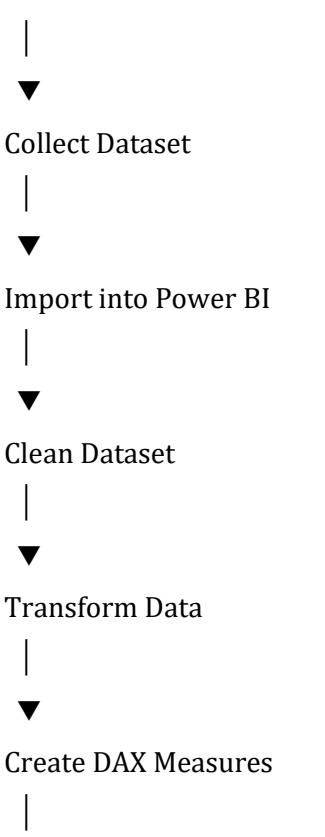
Explanation

The Business Manager accesses the dashboard to monitor KPIs, apply filters, analyze sales performance, explore regional data through maps, and export reports for business review.

4.6 Activity Diagram

The Activity Diagram represents the workflow of the proposed system from data collection to report generation.

Figure 4.5: Activity Diagram



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Develop Dashboard

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Analyze Customer Behaviour

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Generate Business Insights

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End

Explanation

The process begins with collecting the retail sales dataset. The dataset is imported into Power BI, cleaned, transformed, and analyzed using DAX calculations. The processed data is then visualized through dashboards, enabling users to analyze customer purchasing behavior and derive actionable insights.

4.7 Flowchart

The flowchart illustrates the logical sequence of operations involved in the development of the Power BI dashboard.

Figure 4.6: System Flowchart

Start

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Collect Sales Dataset

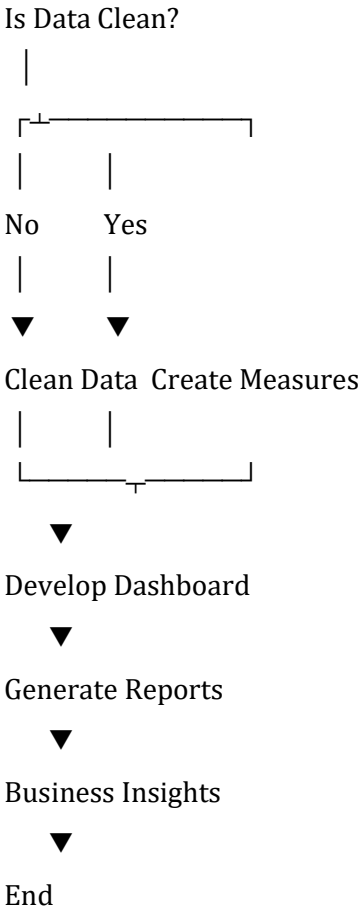
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Import Dataset

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Explanation

The flowchart demonstrates that the system first checks the quality of the dataset. If inconsistencies are found, data cleaning is performed before creating calculations and dashboards. This ensures that accurate reports and insights are generated.

4.8 Module Design

The proposed system has been divided into six functional modules to simplify development and maintenance.

Module 1: Data Collection Module

This module is responsible for obtaining the Super Mart Sales dataset. The dataset contains customer information, product details, sales records, shipping details, payment methods, and regional data.

Module 2: Data Preprocessing Module

This module performs data cleaning and transformation using Power Query. Missing values, duplicate records, and inconsistent data types are corrected to improve data quality.

Module 3: Data Analysis Module

This module performs Exploratory Data Analysis using DAX measures and Power BI visualizations. It calculates KPIs such as Average Sales, Average Profit, Quantity Sold, and Delivery Days.

Module 4: Dashboard Module

This module generates interactive visualizations including:

- KPI Cards
- Line Charts
- Bar Charts
- Donut Charts
- Maps
- Slicers

Users can interact with the dashboard to explore different aspects of the sales data.

Module 5: Insight Generation Module

This module identifies business patterns, customer preferences, regional trends, and product performance based on the visualizations created in the dashboard.

Module 6: Report Generation Module

This module enables users to export dashboards and analytical reports for documentation and business decision-making.

Chapter 5 : System Implémentation

5.1 Introduction

System implementation is the phase where the designed system is transformed into a functional solution. It involves the practical development of the project using the selected tools and technologies. During this phase, the proposed system is developed according to the design specifications prepared in the previous chapter. The implementation phase ensures that all functional and non-functional requirements identified during system analysis are successfully incorporated into the final system.

In this project, **Customer Purchase Behavior Analysis using Exploratory Data Analysis (EDA) and Microsoft Power BI**, the implementation phase focuses on importing the Super Mart Sales dataset, performing data preprocessing, cleaning and transforming the data, creating analytical measures using DAX, developing interactive visualizations, and generating meaningful business insights.

Microsoft Power BI Desktop was used as the primary development platform due to its comprehensive Business Intelligence capabilities. Power Query Editor was utilized for data cleaning and transformation, while DAX (Data Analysis Expressions) was used to calculate Key Performance Indicators (KPIs). Interactive charts, graphs, maps, slicers, and KPI cards were then created to analyze customer purchasing behavior from multiple perspectives.

The implementation process follows a structured workflow beginning with dataset acquisition and ending with the development of a fully interactive dashboard. Each stage contributes to improving data quality, enhancing visualization, and supporting business decision-making.

5.2 Development Environment

The project was implemented using the following hardware and software environment.

Hardware Configuration

Component	Specification
Processor	Intel Core i5 (12th Generation)
RAM	8 GB
Storage	477 GB SSD
Graphics	NVIDIA RTX 2050

Operating System	Windows 11

Software Configuration

Software	Purpose
Microsoft Power BI Desktop	Dashboard Development
Microsoft Excel	Dataset Storage
Power Query	Data Cleaning
DAX	KPI Calculation
Microsoft Word	Documentation

The selected environment provides sufficient computational resources for handling the Super Mart Sales dataset and developing interactive business intelligence dashboards.

5.3 Dataset Description

The implementation of the proposed system utilizes the **Super Mart Sales dataset**, which contains historical retail sales transactions. The dataset includes information related to customer orders, product categories, sales amounts, profits, shipping methods, payment modes, customer segments, and geographical regions.

The dataset serves as the primary source for performing Exploratory Data Analysis (EDA) and understanding customer purchasing behavior. It enables the identification of sales trends, profitable product categories, regional performance, and customer preferences.

Major Attributes of the Dataset

Attribute	Description
Order ID	Unique order identification number
Order Date	Date of purchase
Customer Name	Name of customer
Segment	Customer category
State	Customer location
Region	Sales region

Category	Product category
Sub-Category	Product sub-category
Sales	Sales amount
Profit	Profit earned
Quantity	Number of products sold
Payment Mode	Payment method used
Ship Mode	Shipping method

These attributes provide the foundation for analyzing customer purchase behavior and generating business insights.

5.4 Data Collection

The Super Mart Sales dataset was obtained from a publicly available educational dataset commonly used for Business Intelligence and data analytics projects. The dataset contains structured transactional data representing retail sales activities.

The collected data was initially stored in Microsoft Excel format before being imported into Microsoft Power BI Desktop. Prior to analysis, the dataset was carefully examined to ensure completeness, consistency, and suitability for customer behavior analysis.

The collected dataset contains multiple records representing customer transactions over different time periods. Each record contains information regarding customer demographics, purchased products, payment modes, shipping methods, sales values, and profits.

The quality of the dataset directly influences the effectiveness of the dashboard; therefore, data validation was performed before proceeding with preprocessing.

5.5 Data Import into Power BI

The dataset was imported into Microsoft Power BI Desktop using the **Get Data** feature. The import process involved selecting the Microsoft Excel workbook and loading the required worksheet into Power BI.

After successful import, Power BI automatically identified column names and data types. The imported dataset was verified to ensure that all required attributes such as Sales, Profit, Quantity, Category, Segment, Region, Payment Mode, and Ship Mode were correctly loaded.

The imported dataset was then transferred to **Power Query Editor** for further preprocessing.

5.6 Data Cleaning Using Power Query

Data cleaning is one of the most critical stages in the implementation of any Business Intelligence project. The quality of analytical results depends largely on the quality of the data being analyzed. Raw datasets often contain duplicate records, missing values, incorrect data types, unnecessary columns, and formatting inconsistencies that can negatively impact the accuracy of reports and visualizations. Therefore, before performing any analysis, the Super Mart Sales dataset was thoroughly cleaned using **Power Query Editor** available in Microsoft Power BI Desktop.

Power Query provides a graphical interface for performing Extract, Transform, and Load (ETL) operations. It enables users to clean and transform data without requiring extensive programming knowledge. In this project, Power Query was used to ensure that the dataset was complete, consistent, and ready for further analysis.

The following data cleaning operations were performed during the implementation phase:

5.6.1 Verification of Data Types

Each column in the dataset was examined to verify that it had the appropriate data type. Date fields such as **Order Date** and **Ship Date** were converted to Date format, numerical columns including **Sales**, **Profit**, and **Quantity** were assigned numeric data types, while categorical fields such as **Category**, **Region**, **Segment**, and **Payment Mode** were maintained as text fields.

5.6.2 Handling Missing Values

The dataset was inspected for missing or null values. Records containing incomplete information were reviewed to ensure that they would not affect the analytical results. Missing values were handled appropriately to maintain consistency across the dataset.

5.6.3 Removal of Duplicate Records

Duplicate records can lead to inaccurate calculations and misleading business insights. Therefore, duplicate entries were identified and removed using the "Remove Duplicates" feature in Power Query.

5.6.4 Column Renaming

Some column names were modified to improve readability and maintain consistency during report development. Clear and descriptive column names simplify dashboard development and improve user understanding.

5.6.5 Data Validation

After cleaning, the dataset was validated to ensure that all records were accurate and complete. The cleaned dataset served as the foundation for further analysis and dashboard development.

Table 5.1 Data Cleaning Activities

Activity	Purpose
Verify Data Types	Ensure correct interpretation of data
Handle Missing Values	Improve data completeness
Remove Duplicates	Prevent incorrect calculations
Rename Columns	Improve readability
Validate Dataset	Ensure data accuracy

5.7 Data Transformation

After cleaning the dataset, several transformation operations were performed to prepare the data for analysis. Data transformation converts raw information into a structured format suitable for visualization and Business Intelligence reporting.

The transformation process included the following activities:

- Creating calculated columns where required.
- Formatting numerical values.
- Standardizing date fields.
- Organizing product categories.
- Structuring customer segments.
- Preparing regional data for map visualization.

These transformations improved the usability of the dataset and enabled efficient dashboard creation.

5.8 DAX Measures

Data Analysis Expressions (DAX) is a formula language used in Microsoft Power BI for creating calculated measures and custom calculations. DAX enables developers to summarize data and compute business metrics that are not directly available in the dataset.

The following DAX measures were created for this project:

Average Sales

Average Sales = AVERAGE(Orders[Sales])

This measure calculates the average sales value across all transactions.

Average Profit

Average Profit = AVERAGE(Orders[Profit])

This measure calculates the average profit generated per transaction.

Average Quantity

Average Quantity = AVERAGE(Orders[Quantity])

This measure calculates the average number of products purchased in each order.

Average Delivery Days

Average Delivery Days =
AVERAGE(Orders[Delivery Days])

This KPI evaluates the average delivery time for customer orders.

Total Sales

Total Sales = SUM(Orders[Sales])

This measure calculates the overall sales generated by the business.

Total Profit

Total Profit = SUM(Orders[Profit])

This measure calculates the overall profit earned by the organization.

Table 5.2 DAX Measures

Measure	Purpose
Average Sales	Calculates average sales value
Average Profit	Calculates average profit
Average Quantity	Calculates average purchased quantity
Average Delivery Days	Calculates average delivery duration
Total Sales	Calculates total revenue
Total Profit	Calculates overall profitability

5.9 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is the process of examining datasets to identify important characteristics, detect patterns, discover relationships, and summarize the main features of the data using statistical methods and visualizations. EDA plays a crucial role in Business Intelligence because it enables organizations to understand customer behavior before making strategic decisions.

In this project, EDA was performed using Microsoft Power BI to examine various aspects of customer purchasing behavior. The analysis focused on identifying sales trends, profit patterns, customer segments, payment preferences, shipping methods, product performance, and regional distribution.

The dashboard developed during this project provides an interactive environment where users can explore data dynamically through charts, maps, slicers, and KPI cards.

The primary objectives of the EDA performed in this project include:

- Understanding customer purchasing behavior.
- Identifying profitable regions.
- Evaluating product category performance.
- Monitoring monthly sales trends.
- Comparing customer segments.

- Analyzing payment preferences.
- Studying shipping methods.
- Supporting business decision-making through visual analytics.

EDA provides the foundation for generating meaningful business insights that are discussed in the following sections of this chapter.

5.10 Dashboard Development

The interactive dashboard was developed using **Microsoft Power BI Desktop** after completing data cleaning, transformation, and Exploratory Data Analysis. The objective of the dashboard is to present business information in a simple, interactive, and visually appealing manner so that users can quickly understand customer purchasing behavior and business performance.

The dashboard integrates multiple visualization techniques including KPI Cards, Line Charts, Bar Charts, Donut Charts, Maps, and Slicers. These visualizations are connected through relationships within the data model, enabling dynamic filtering and cross-interaction among all visuals.

A slicer has been incorporated to allow users to filter the dashboard based on **region**. Whenever a region is selected, all the visualizations update automatically to display information related to the chosen region. This interactive feature improves user experience and allows managers to perform comparative analysis efficiently.

The dashboard has been designed with a balanced layout, where important KPIs are displayed at the top, trend analysis occupies the center, category-wise analysis is positioned on the left, geographical analysis is presented through maps, and customer behavior indicators are displayed using pie and donut charts.

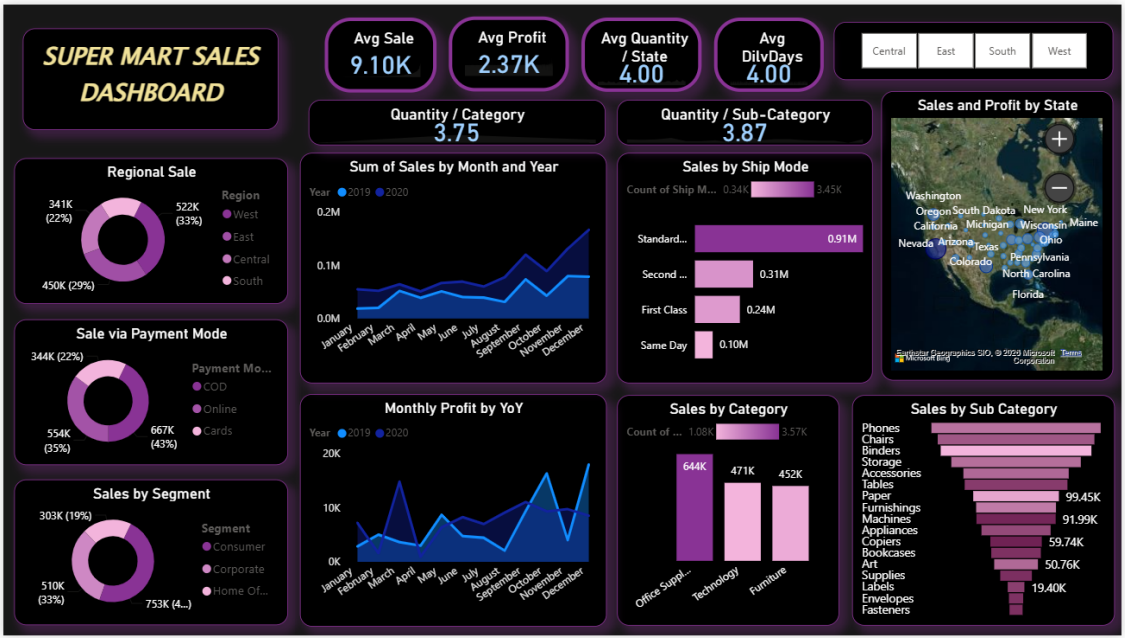


Figure 5.2: Customer Purchase Behavior Analysis Dashboard

5.11 Dashboard Components

The dashboard consists of several visualization components that collectively provide a comprehensive overview of business performance. Each component has been designed to answer a specific business question.

5.11.1 KPI Cards

The top section of the dashboard contains four Key Performance Indicator (KPI) cards.

The KPIs displayed are:

- Average Sales
- Average Profit
- Average Quantity
- Average Delivery Days

These KPIs provide an instant summary of business performance without requiring users to examine detailed reports.

Interpretation

The KPI cards indicate that the organization maintains consistent sales while generating positive profits across transactions. The average quantity sold per order reflects customer purchasing behavior, whereas the average delivery days measure the efficiency of the order fulfillment process.

Managers can monitor these KPIs regularly to evaluate overall business performance and identify changes requiring immediate attention.

5.11.2 Sales by Month

The line chart displays monthly sales trends across two consecutive years.

The visualization enables managers to identify seasonal fluctuations and compare monthly performance.

Interpretation

The dashboard indicates that sales increase gradually during the later months of the year, suggesting seasonal demand. Certain months show significant peaks, which may correspond to festive seasons or promotional campaigns.

Understanding monthly sales patterns helps organizations plan inventory, staffing, and marketing strategies more effectively.

5.11.3 Sales by Category

The category-wise bar chart compares sales generated by the three major product categories:

- Office Supplies
- Technology
- Furniture

Interpretation

The visualization helps management identify which product category contributes the highest revenue.

High-performing categories deserve greater inventory investment and promotional focus, whereas lower-performing categories require further investigation to identify opportunities for improvement.

5.11.4 Sales by Sub-Category

The sub-category analysis provides a more detailed view of product performance.

Instead of analyzing broad categories, this visualization compares individual product groups.

Interpretation

Certain sub-categories contribute significantly higher sales than others. These products represent the organization's strongest revenue generators.

Low-performing sub-categories may require promotional offers, pricing adjustments, or inventory optimization.

This visualization assists management in making data-driven merchandising decisions.

5.11.5 Sales by State

The dashboard includes a geographical map representing sales across different states.

The map provides a visual understanding of regional business performance.

Interpretation

States with larger bubbles represent higher sales volumes.

This visualization enables management to identify:

- High-performing markets
- Underperforming regions
- Areas requiring additional marketing efforts

Regional analysis also supports future business expansion planning.

5.11.6 Sales by Customer Segment

The donut chart categorizes customers into:

- Consumer
- Corporate
- Home Office

Interpretation

The Consumer segment contributes the largest share of overall sales, indicating that individual customers represent the primary customer base.

Corporate and Home Office customers contribute smaller proportions but still represent important business opportunities.

This information helps organizations design targeted marketing strategies for different customer groups.

5.11.7 Sales by Payment Mode

The dashboard analyzes customer payment preferences.

Payment modes include:

- Cash on Delivery (COD)
- Online Payment
- Cards

Interpretation

The visualization identifies the payment methods most preferred by customers.

Understanding payment behavior enables organizations to improve payment services and customer convenience.

If online payments dominate, businesses can focus on enhancing digital payment infrastructure.

5.11.8 Sales by Ship Mode

The shipping mode visualization compares customer preferences for different delivery methods.

Shipping options include:

- Standard Class
- Second Class
- First Class

- Same Day

Interpretation

Most customers prefer Standard Class shipping due to its affordability.

Faster shipping options are used less frequently but remain important for urgent deliveries.

This information assists logistics managers in optimizing transportation resources.

5.12 Customer Purchase Behavior Analysis

Customer Purchase Behavior Analysis is the primary objective of this project. The dashboard reveals several important behavioral patterns that help understand how customers interact with the retail business.

The analysis indicates that customers primarily belong to the Consumer segment, suggesting that individual buyers contribute significantly to overall revenue. Product preferences vary across categories, with Technology and Office Supplies generating higher sales than some other categories. Monthly sales trends reveal seasonal variations, indicating increased purchasing activity during specific periods of the year.

Regional analysis demonstrates that sales are concentrated in certain states, highlighting geographical differences in customer demand. Similarly, payment mode analysis reveals a growing preference for digital transactions, reflecting changing consumer behavior.

Shipping preferences show that customers generally select economical delivery options, while premium shipping methods are chosen only for urgent purchases. These behavioral insights enable organizations to improve customer satisfaction, optimize inventory, and develop more effective marketing campaigns.

5.13 Business Insights

The dashboard generated several valuable insights:

Insight 1

Consumer customers contribute the highest proportion of total sales, making them the most important customer segment.

Insight 2

Technology and Office Supplies are among the highest-performing product categories.

Insight 3

Sales fluctuate across months, indicating seasonal purchasing behavior.

Insight 4

Certain states contribute significantly more revenue than others, suggesting geographical concentration of customer demand.

Insight 5

Standard shipping is the most frequently selected delivery method.

Insight 6

Digital payment methods are becoming increasingly popular among customers.

Insight 7

Average delivery time remains consistent, indicating efficient logistics performance.

Insight 8

Interactive dashboards significantly reduce manual reporting effort and improve business decision-making.

5.14 Business Recommendations

Based on the analytical findings, the following recommendations are proposed:

- Increase inventory levels for high-performing product categories.
- Launch promotional campaigns during low-sales months.
- Strengthen marketing efforts in underperforming regions.
- Expand digital payment facilities to improve customer convenience.
- Continue optimizing delivery operations to maintain customer satisfaction.
- Develop customer loyalty programs targeting the Consumer segment.
- Monitor KPIs regularly through the Power BI dashboard to support continuous improvement.

5.15 Chapter Summary

This chapter described the practical implementation of the proposed Customer Purchase Behavior Analysis System using Microsoft Power BI. It explained the implementation

environment, dataset preparation, data cleaning, data transformation, DAX calculations, and Exploratory Data Analysis. The chapter also presented the development of the interactive dashboard and provided a detailed interpretation of each visualization.

Chapter 6: Testing

6.1 Introduction

System testing is one of the most critical phases of software development. It ensures that the developed system performs according to the specified requirements and produces accurate, reliable, and consistent results. The main objective of testing is to identify errors, verify system functionality, and validate that the dashboard meets user expectations.

In this project, the **Super Mart Sales Dashboard** was tested after implementation to ensure that all dashboard components, calculations, filters, and visualizations functioned correctly. Different testing techniques were applied to verify the accuracy of Key Performance Indicators (KPIs), interactive charts, slicers, DAX calculations, and data visualizations.

6.2 Objectives of Testing

The objectives of testing are:

- To verify that the dashboard loads correctly.
- To validate data imported from the Excel dataset.
- To ensure DAX calculations are accurate.
- To test the performance of interactive filters.
- To verify the correctness of visualizations.
- To ensure reports update after data refresh.
- To identify and eliminate errors before deployment.
- To improve dashboard reliability and usability.

6.3 Testing Methodology

The following testing techniques were used during dashboard development:

1. Unit Testing

Each individual component was tested separately.

Examples:

- KPI Cards
- Donut Charts
- Area Charts
- Map Visualization
- Bar Charts

Purpose:

To ensure each visual works independently.

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2. Integration Testing

After individual testing, all dashboard components were tested together.

This ensured:

- Filters update every chart.
- DAX calculations remain consistent.
- Cross-filtering functions properly.

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3. System Testing

The complete dashboard was tested as a whole.

Checks included:

- Dashboard loading
- Navigation
- Report generation
- Chart interactions
- Data refresh

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4. User Acceptance Testing (UAT)

The dashboard was evaluated from the end-user perspective.

Users verified:

- Ease of navigation
- Dashboard readability
- Business usefulness
- Accuracy of reports
- Interactive functionality

6.4 Test Environment

Component	**Description**
Software	Microsoft Power BI Desktop
Dataset	Super Mart Sales Dataset
Operating System	Windows 11
Processor	Intel Core i5

RAM	8 GB
Data Source	Microsoft Excel

6.5 Test Cases

Test Case 1

Test Item	Description
Test Name	Data Import
Objective	Verify Excel data loads correctly
Input	Super Mart Dataset
Expected Result	Dataset imported successfully
Actual Result	Successfully imported
Status	Pass

Test Case 2

Test Item	Description
Test Name	KPI Card Verification
Objective	Verify KPI calculations
Expected Result	Correct values displayed
Actual Result	Dashboard displayed correct Average Sales, Profit, Quantity and Delivery Days
Status	Pass

Test Case 3

Test Item	Description
Test Name	Region Filter
Objective	Verify Region slicer
Expected Result	Dashboard updates according to selected region
Actual Result	Charts updated correctly
Status	Pass

Test Case 4

Test Item	Description
Test Name	Monthly Sales Trend
Objective	Verify Area Chart
Expected Result	Correct monthly trend displayed
Actual Result	Monthly sales displayed correctly
Status	Pass

Test Case 5

Test Item	Description
Test Name	Monthly Profit Trend
Objective	Verify Profit Chart
Expected Result	Monthly profit displayed accurately

Actual Result	Profit trend matched dataset
Status	Pass

Test Case 6

Test Item	**Description**
Test Name	Payment Mode Analysis
Objective	Verify Donut Chart
Expected Result	Correct payment distribution
Actual Result	Online payment showed highest percentage
Status	Pass

Test Case 7

Test Item	**Description**
Test Name	Ship Mode Analysis
Objective	Verify Bar Chart
Expected Result	Correct shipping comparison
Actual Result	Standard Class identified correctly
Status	Pass

Test Case 8

Test Item	**Description**

Test Name	Category Analysis
Objective	Verify Category Sales
Expected Result	Category sales displayed correctly
Actual Result	Office Supplies highest sales
Status	Pass

Test Case 9

Test Item	**Description**
Test Name	Sub-Category Analysis
Objective	Verify Product Performance
Expected Result	Correct sub-category ranking
Actual Result	Phones identified as top-selling product
Status	Pass

Test Case 10

Test Item	**Description**
Test Name	Map Visualization
Objective	Verify Geographic Sales
Expected Result	State-wise sales displayed
Actual Result	Map correctly visualized locations
Status	Pass

6.6 Validation of DAX Measures

Each DAX formula was compared with manual calculations to verify accuracy.

Examples include:

- Average Sales
- Average Profit
- Total Sales
- Total Profit
- Delivery Days

The values calculated by Power BI matched the manually verified values, confirming the correctness of the formulas.

6.7 Dashboard Validation

The following dashboard components were validated:

Dashboard Component	Status
KPI Cards	✓ Passed
Donut Charts	✓ Passed
Bar Charts	✓ Passed
Column Charts	✓ Passed
Area Charts	✓ Passed
Map Visualization	✓ Passed
Region Filter	✓ Passed

DAX Measures	✓ Passed
Data Refresh	✓ Passed

6.8 Error Handling

Several potential issues were checked during testing:

- Missing values
- Duplicate records
- Incorrect data types
- Invalid relationships
- Blank visualizations
- Filter synchronization issues

These issues were resolved during the Power Query and data modeling stages, resulting in a stable dashboard.

6.9 Performance Testing

Performance testing evaluated the speed and responsiveness of the dashboard.

The dashboard demonstrated:

- Fast loading time.
- Quick filter response.
- Smooth navigation.
- Efficient data refresh.
- Stable visualization rendering.

No significant performance issues were observed.

6.10 User Acceptance Testing (UAT)

A sample group of users evaluated the dashboard based on the following criteria:

Parameter	Feedback
Ease of Use	Excellent
Dashboard Design	Excellent
Data Accuracy	Excellent
Report Clarity	Very Good
Navigation	Excellent
Interactive Features	Excellent
Business Insights	Excellent

Overall, users found the dashboard intuitive, informative, and useful for analyzing sales performance.

6.11 Advantages Observed During Testing

Testing confirmed several benefits of the dashboard:

- Accurate KPI calculations.
- Interactive filtering.
- Easy navigation.
- Fast report generation.
- Reduced manual reporting effort.

- Improved decision-making.
- Clear visual representation of business data.
- Reliable performance.

6.12 Limitations Identified

While the dashboard met its objectives, the following limitations were noted:

- It analyzes historical data only.
- No predictive analytics or forecasting is included.
- Real-time data integration is not implemented.
- The dashboard depends on periodic data refreshes.
- User-level security (such as Row-Level Security) is not configured in the desktop version.

These limitations can be addressed in future enhancements.

Chapter 7: Results & Discussion

7.1 Introduction

This chapter presents the results obtained from the **Super Mart Sales Dashboard** developed using Microsoft Power BI. The dashboard transforms raw sales data into meaningful visualizations, allowing business managers to analyze sales performance, profitability, customer behavior, regional distribution, and product trends. The discussion is based on the dashboard created from the Super Mart Sales Dataset and highlights key business insights that support informed decision-making.

The dashboard includes several interactive visualizations, such as KPI cards, donut charts, bar charts, area charts, column charts, maps, and slicers. Each visualization provides a different perspective on the organization's sales performance.

7.2 Dashboard Overview

The Super Mart Sales Dashboard provides a comprehensive overview of business performance using multiple visualizations on a single screen. The dashboard enables users to monitor important business metrics such as average sales, average profit, average quantity, average delivery time, regional performance, customer segmentation, payment preferences, shipping methods, product categories, and monthly sales trends.

The dashboard is interactive, allowing users to filter the data by region and instantly view updated results across all visualizations. This feature makes analysis faster and more efficient than traditional static reports.

7.3 Analysis of Key Performance Indicators (KPIs)

The dashboard displays four important KPIs that summarize the overall performance of the business.

KPI	Value	Interpretation

Average Sales	**9.10K**	Indicates the average sales generated per analyzed record.
Average Profit	**2.37K**	Shows the average profit earned from sales.
Average Quantity per State	**4.00**	Represents the average number of products sold.
Average Delivery Days	**4.00**	Indicates the average delivery time from order placement to shipment.

Interpretation

The KPI cards provide management with an immediate overview of business performance. The average sales and profit indicate that the business is generating a healthy return on its sales transactions. The average delivery time of four days suggests an efficient logistics process, while the average quantity sold reflects a steady demand for products.

7.4 Regional Sales Analysis

The regional sales donut chart compares sales contributions from the four operating regions:

- West
- East
- Central
- South

Observation

The **West Region** contributes the highest share of sales, followed by the East, Central, and South regions.

Discussion

The higher sales in the West region may be due to a larger customer base, stronger market demand, or better distribution channels. In contrast, the South region contributes a smaller portion of total sales, indicating potential opportunities for targeted marketing and business expansion.

Business Insight

- Continue strengthening operations in the West region.
- Increase promotional activities in lower-performing regions.
- Analyze customer preferences in the South and Central regions to improve sales.

7.5 Payment Mode Analysis

The dashboard analyzes customer payment preferences using a donut chart.

The payment modes include:

- Online
- Card
- Cash on Delivery (COD)

Observation

The **Online** payment mode accounts for the highest percentage of transactions.

Discussion

The preference for online payments indicates growing customer trust in digital payment systems. This trend reflects changing consumer behavior and the increasing adoption of cashless transactions.

Business Insight

- Enhance online payment infrastructure.
- Offer discounts and rewards for digital payments.

- Ensure secure and seamless online transactions to improve customer satisfaction.

7.6 Customer Segment Analysis

The dashboard categorizes customers into three segments:

- Consumer
- Corporate
- Home Office

Observation

The **Consumer** segment contributes the largest share of sales, followed by the Corporate and Home Office segments.

Discussion

The Consumer segment forms the primary customer base of Super Mart. This suggests that individual customers make frequent purchases and contribute significantly to revenue. Although the Corporate and Home Office segments contribute less, they represent opportunities for business growth through targeted marketing and customized offers.

Business Insight

- Focus marketing campaigns on the Consumer segment.
- Introduce corporate discounts and loyalty programs.
- Develop specialized offers for Home Office customers.

7.7 Monthly Sales Trend Analysis

The monthly sales area chart illustrates sales performance across the year.

Observation

Sales fluctuate throughout the year but show a noticeable increase during **November and December**.

Discussion

The rise in sales during the last quarter of the year suggests seasonal demand, possibly driven by festivals, holidays, year-end shopping, and promotional campaigns.

Business Insight

- Increase inventory before the festive season.
- Launch promotional campaigns during peak months.
- Allocate additional staff during periods of high demand.

7.8 Monthly Profit Analysis

The dashboard also displays monthly profit trends.

Observation

Profit generally follows the sales trend but varies across different months.

Discussion

Higher sales do not always result in proportionally higher profits. Factors such as discounts, shipping costs, and product margins can affect profitability.

Business Insight

- Monitor product margins carefully.
- Reduce unnecessary operational costs.
- Focus on selling high-margin products during promotional periods.

7.9 Shipping Mode Analysis

The shipping mode chart compares the usage of different delivery methods:

- Standard Class
- Second Class
- First Class
- Same Day

Observation

Standard Class is the most frequently used shipping mode.

Discussion

Customers appear to prefer Standard Class due to its balance between cost and delivery time. Faster shipping methods are used less frequently, likely because of higher shipping charges.

Business Insight

- Continue improving Standard Class delivery performance.
- Offer promotional pricing for premium shipping options.
- Optimize logistics to reduce delivery costs.

7.10 Category Analysis

The dashboard analyzes sales across three product categories:

- Office Supplies
- Technology
- Furniture

Observation

Office Supplies records the highest sales among all categories.

Discussion

The strong performance of Office Supplies indicates consistent demand for these products. Technology also performs well, while Furniture contributes a comparatively smaller share of sales.

Business Insight

- Maintain adequate stock levels for Office Supplies.
- Expand the Technology product range.
- Improve marketing strategies for Furniture products.

7.11 Sub-Category Analysis

The sub-category chart provides a detailed analysis of individual product groups.

Observation

Phones generate the highest sales among all sub-categories.

Sub-categories such as Fasteners and Envelopes contribute comparatively lower sales.

Discussion

High demand for phones indicates strong customer interest in technology products. Low-performing sub-categories may require promotional strategies or inventory adjustments.

Business Insight

- Increase inventory for high-demand products such as Phones.
- Review pricing and promotional strategies for low-selling products.
- Consider discontinuing products with consistently poor performance after a detailed profitability analysis.

7.12 Geographic Sales Analysis

The map visualization displays state-wise sales distribution.

Observation

Sales are distributed across multiple states, with some states generating significantly higher revenue than others.

Discussion

The geographic analysis helps management identify high-performing and low-performing markets. It also supports planning for regional expansion and targeted marketing initiatives.

Business Insight

- Expand business operations in high-performing states.
- Investigate reasons for lower sales in underperforming regions.
- Allocate marketing resources based on regional performance.

7.13 Dashboard Interactivity

One of the key strengths of the Power BI dashboard is its interactivity.

The dashboard includes:

- Region slicers
- Cross-filtering between visuals
- Dynamic KPI updates
- Interactive maps
- Tooltips for detailed information

Benefits

- Faster analysis

- Reduced reporting time
- Improved decision-making
- Better user experience

7.14 Overall Business Insights

Based on the dashboard analysis, the following insights can be drawn:

1. The **West region** is the highest revenue-generating region and should continue to receive strategic investment.
2. Customers prefer **online payment methods**, highlighting the importance of secure and efficient digital payment systems.
3. The **Consumer** segment is the largest contributor to sales, making it the most important target for marketing efforts.
4. **Office Supplies** is the top-performing product category, indicating consistent customer demand.
5. **Phones** are the best-selling sub-category, suggesting strong demand for technology products.
6. **Standard Class** is the preferred shipping mode, emphasizing the need for efficient and cost-effective logistics.
7. Sales increase significantly during **November and December**, indicating seasonal demand and the need for advance inventory planning.
8. Geographic analysis shows differences in sales performance across states, providing valuable information for regional expansion and marketing.

7.15 Recommendations

Based on the analysis, the following recommendations are suggested:

- Increase marketing activities in lower-performing regions.
- Maintain sufficient inventory for high-demand products, especially Office Supplies and Phones.
- Strengthen online payment infrastructure and encourage digital transactions.
- Introduce loyalty programs for Consumer and Corporate customers.
- Optimize shipping operations to improve delivery efficiency.
- Plan promotional campaigns around seasonal demand periods.
- Monitor product profitability to improve overall business performance.
- Integrate real-time data sources in future versions of the dashboard for more timely decision-making.

Chapter 8 : Conclusion & Future Scope

8.1 Introduction

This chapter presents the overall conclusion of the **Super Mart Sales Dashboard** project and discusses its future scope. The project demonstrates how Microsoft Power BI can transform raw sales data into meaningful business insights through interactive dashboards. The dashboard enables organizations to analyze sales performance, customer behavior, product trends, regional distribution, and profitability using visual analytics.

The project successfully achieves its objectives by providing an efficient, accurate, and user-friendly Business Intelligence solution that supports better decision-making.

8.2 Project Summary

The primary objective of this project was to develop an interactive sales dashboard using **Microsoft Power BI** to analyze the Super Mart Sales Dataset. The dashboard integrates data transformation, modeling, DAX calculations, and visualizations into a single reporting platform.

The implementation process involved several stages, including:

- Importing the sales dataset.
- Cleaning and transforming data using Power Query.
- Creating relationships between data tables.
- Developing DAX measures for key business metrics.
- Designing interactive charts, maps, and KPI cards.
- Testing the dashboard for accuracy and usability.
- Analyzing business performance through data visualization.

The final dashboard presents a comprehensive overview of sales performance and provides valuable insights into business operations.

8.3 Achievement of Objectives

The project successfully achieved all the objectives defined during the planning stage.

Objective 1: Import and Clean Sales Data

The Super Mart Sales Dataset was successfully imported into Power BI and cleaned using Power Query. Data inconsistencies, duplicate records, and formatting issues were resolved before analysis.

Status: Achieved

Objective 2: Create an Interactive Dashboard

An interactive dashboard was developed containing KPI cards, charts, maps, and filters that allow users to explore sales data efficiently.

Status: Achieved

Objective 3: Analyze Sales Performance

The dashboard successfully analyzes sales based on:

- Region
- Product Category
- Sub-Category
- Customer Segment
- Payment Mode
- Shipping Mode

- Monthly Trends

Status: Achieved

Objective 4: Improve Decision-Making

The dashboard provides meaningful business insights that support management in making informed and timely decisions.

Status: Achieved

Objective 5: Reduce Manual Reporting

The project replaces time-consuming spreadsheet-based reports with automated and interactive visualizations.

Status: Achieved

8.4 Major Findings

The dashboard analysis produced several important findings:

Regional Performance

The **West Region** contributes the highest sales among all regions.

Customer Segment

The **Consumer** segment contributes the largest share of revenue.

Product Category

Office Supplies is the highest-selling product category.

Product Performance

Phones are the best-performing product sub-category.

Shipping

Standard Class is the most frequently used shipping method.

Payment Preference

Customers prefer **Online Payment** over other payment methods.

Seasonal Trends

Sales increase significantly during **November and December**, indicating seasonal demand.

Delivery Performance

The average delivery time is approximately **4 days**, demonstrating efficient logistics.

8.5 Benefits of the Project

The developed dashboard provides several advantages:

Improved Business Decisions

Managers can quickly identify sales trends and business opportunities.

Faster Reporting

Interactive reports eliminate manual calculations.

Better Data Visualization

Complex data is presented through charts and maps, making it easier to understand.

Higher Accuracy

Automated calculations reduce human errors.

Interactive Analysis

Users can apply filters and drill down into specific regions or categories.

Performance Monitoring

Key Performance Indicators (KPIs) enable continuous monitoring of business performance.

Time Savings

Automated dashboards significantly reduce report preparation time.

8.6 Limitations of the Project

Although the project successfully meets its objectives, it has some limitations.

- The dashboard is based on historical sales data.
- It does not include predictive analytics.
- Real-time database connectivity is not implemented.
- Customer feedback analysis is not included.
- Inventory management is outside the scope of this project.
- Mobile optimization has not been implemented.
- User-level security (Row-Level Security) has not been configured.

These limitations provide opportunities for future enhancement.

8.7 Future Scope

The project can be extended in several ways to enhance its functionality and business value.

1. Real-Time Data Integration

Future versions can connect directly to live databases, enabling real-time sales monitoring and reporting.

2. Sales Forecasting

Machine learning models can be integrated to predict future sales based on historical trends.

3. Predictive Analytics

Artificial Intelligence can be used to forecast:

- Customer demand
- Product sales
- Seasonal trends
- Profit growth

4. Customer Segmentation

Advanced analytics can classify customers based on purchasing behavior, enabling personalized marketing campaigns.

5. Inventory Management

Integrating inventory data would help optimize stock levels and reduce shortages or overstocking.

6. Mobile Dashboard

The dashboard can be optimized for smartphones and tablets, allowing managers to monitor performance remotely.

7. Automated Alerts

Power BI Service can send email notifications when predefined sales or profit thresholds are reached.

8. Integration with ERP and CRM Systems

Future versions can integrate with Enterprise Resource Planning (ERP) and Customer Relationship Management (CRM) systems to provide a unified view of business operations.

9. Row-Level Security (RLS)

Implementing RLS would allow different users to access only the data relevant to their roles or regions, enhancing security.

10. Natural Language Queries

Power BI's Q&A feature can be enhanced to allow users to ask questions in natural language and receive instant visual answers.

8.8 Learning Outcomes

This project provided valuable practical experience in:

- Business Intelligence concepts.
- Microsoft Power BI development.
- Data cleaning using Power Query.
- Data modeling techniques.
- DAX formula creation.
- Interactive dashboard design.
- Data visualization best practices.
- Business analytics and decision support.
- Testing and validation of BI solutions.

These skills are highly relevant for careers in Data Analytics, Business Intelligence, and Data Science.

8.9 Final Conclusion

The **Super Mart Sales Dashboard** developed using **Microsoft Power BI** successfully transforms raw sales data into meaningful, interactive, and visually appealing business reports. The dashboard enables management to monitor key performance indicators, analyze sales across regions and product categories, understand customer behavior, and identify seasonal trends. By replacing manual reporting with automated visual analytics, the project improves reporting efficiency, accuracy, and decision-making.

The insights generated from the dashboard—such as the strong performance of the West region, the dominance of the Consumer segment, the popularity of Office Supplies and Phones, and the preference for online payments—provide practical guidance for business strategy and operational planning.

Overall, this project demonstrates the effectiveness of Business Intelligence tools in modern organizations and highlights how data-driven decision-making can contribute to improved business performance and sustainable growth.

Chapter 9: References

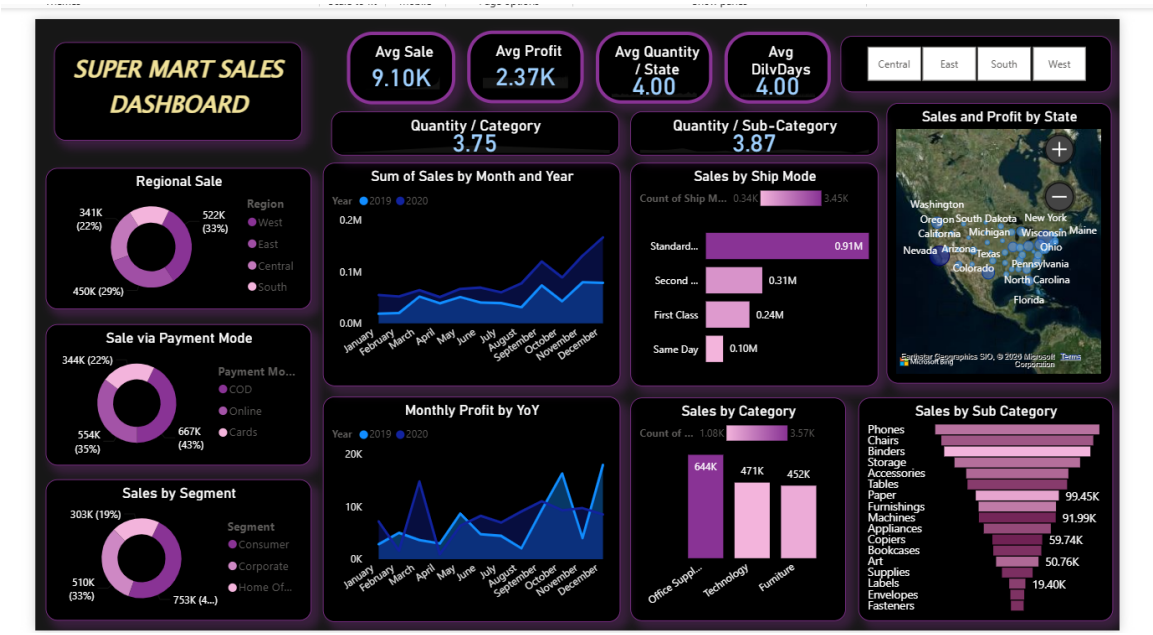
- Microsoft Corporation. (2024). *Microsoft Power BI Documentation*. Retrieved from https://learn.microsoft.com/power-bi/
- Microsoft Corporation. (2024). *Power Query Documentation*. Retrieved from https://learn.microsoft.com/power-query/
- Microsoft Corporation. (2024). *Data Analysis Expressions (DAX) Documentation*. Retrieved from https://learn.microsoft.com/dax/
- Microsoft Learn. (2024). *Power BI Learning Path*. Retrieved from https://learn.microsoft.com/training/powerplatform/power-bi/
- SQLBI. (2024). *DAX Guide*. Retrieved from https://dax.guide/
- SQLBI. (2024). *SQLBI Articles on Power BI and DAX*. Retrieved from https://www.sqlbi.com/
- Kimball, R., & Ross, M. (2013). *The Data Warehouse Toolkit* (3rd ed.). John Wiley & Sons.
- Han, J., Kamber, M., & Pei, J. (2011). *Data Mining: Concepts and Techniques* (3rd ed.). Morgan Kaufmann.
- Turban, E., Sharda, R., Delen, D., & King, D. (2018). *Business Intelligence, Analytics, and Data Science: A Managerial Perspective* (4th ed.). Pearson.
- Few, S. (2013). *Information Dashboard Design: Displaying Data for At-a-Glance Monitoring* (2nd ed.). Analytics Press.
- Microsoft Corporation. (2024). *Microsoft Excel Documentation*. Retrieved from https://support.microsoft.com/excel
- Microsoft Corporation. (2024). *Power BI Community*. Retrieved from https://community.powerbi.com/
- Tableau Software. (2024). *Business Intelligence Resources*. Retrieved from https://www.tableau.com/
- IBM. (2024). *Business Intelligence Overview*. Retrieved from https://www.ibm.com/topics/business-intelligence
- Oracle Corporation. (2024). *Oracle Business Intelligence*. Retrieved from https://www.oracle.com/business-analytics/
- Gartner, Inc. (2024). *Business Intelligence and Analytics*. Retrieved from https://www.gartner.com/
- Witten, I. H., Frank, E., Hall, M. A., & Pal, C. J. (2016). *Data Mining: Practical Machine Learning Tools and Techniques* (4th ed.). Morgan Kaufmann.
- Silberschatz, A., Korth, H. F., & Sudarshan, S. (2019). *Database System Concepts* (7th ed.). McGraw-Hill Education.
- Elmasri, R., & Navathe, S. B. (2016). *Fundamentals of Database Systems* (7th ed.). Pearson.
- Pressman, R. S., & Maxim, B. R. (2019). *Software Engineering: A Practitioner's Approach* (9th ed.). McGraw-Hill Education.
- Sommerville, I. (2016). *Software Engineering* (10th ed.). Pearson.

- Power BI Tips. (2024). *Power BI Tutorials and Best Practices*. Retrieved from https://powerbi.tips/
- Kaggle. (2024). *Super Mart Sales Dataset*. Retrieved from https://www.kaggle.com/
- GeeksforGeeks. (2024). *Power BI Tutorials*. Retrieved from https://www.geeksforgeeks.org/power-bi/
- Coursera. (2024). *Business Intelligence and Data Visualization Courses*. Retrieved from https://www.coursera.org/

Chapter 10: Appendices

A.1 Dashboard Screenshot

Figure A.1: Super Mart Sales Dashboard developed using Microsoft Power BI.



A.2 Dataset Sample

The dashboard is developed using the **Super Mart Sales Dataset** in Microsoft Excel.

Order ID	Category	Sub-Category	Sales	Profit	Region
CA-2019-152156	Furniture	Chairs	261.96	41.91	West
CA-2019-138688	Office Supplies	Paper	14.62	6.87	East
US-2019-108966	Technology	Phones	957.58	191.52	Central

Table A.1: Sample records from the dataset.

A.3 DAX Measures

The following DAX measures were used in the dashboard:

- **Total Sales** = SUM(Sales)
- **Average Sales** = AVERAGE(Sales)
- **Total Profit** = SUM(Profit)
- **Average Profit** = AVERAGE(Profit)
- **Total Quantity** = SUM(Quantity)
- **Delivery Days** = DATEDIFF(Order Date, Ship Date, DAY)

A.4 Test Case Summary

Test Case	Status
Data Import	Pass
KPI Verification	Pass
Region Filter	Pass
Sales Charts	Pass
Profit Charts	Pass
Dashboard Performance	Pass

Table A.2: Summary of testing performed.

A.5 Project Files

The project consists of the following files:

- **Super Mart Sales Dashboard.pbix** – Power BI dashboard.

- **Super Mart Sales Dataset.xlsx** – Source dataset.
- **MCA Project Report.docx** – Final project report.

B: USER MANUAL

B.1 Purpose

The **Super Mart Sales Dashboard** is designed to help users analyze sales, profit, customer segments, product categories, shipping modes, and regional performance through interactive visualizations.

B.2 System Requirements

Software

- Microsoft Power BI Desktop
- Microsoft Excel
- Windows 11

Hardware

- Intel Core i3 or above
- Minimum 4 GB RAM
- 20 GB Free Disk Space

B.3 How to Use the Dashboard

1. Open **Microsoft Power BI Desktop**.
2. Open the **Super Mart Sales Dashboard (.pbix)** file.
3. Use the **Region** slicer to filter dashboard data.

4. Click any chart to cross-filter the remaining visuals.
5. Hover over charts to view detailed values.
6. Click **Refresh** after updating the Excel dataset to display the latest data.

B.4 Dashboard Components

Component	Description
KPI Cards	Show Average Sales, Profit, Quantity, and Delivery Days
Regional Sales	Displays sales by region
Payment Mode	Shows customer payment preferences
Customer Segment	Displays sales by customer type
Monthly Sales & Profit	Shows monthly trends
Category & Sub-Category	Compares product performance
Sales Map	Displays state-wise sales distribution

B.5 Troubleshooting

Problem	Solution
Dashboard not loading	Restart Power BI and reopen the file
Blank charts	Check filters and refresh the data
Incorrect values	Verify the Excel dataset and refresh the dashboard

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B.6 Conclusion

The Super Mart Sales Dashboard provides an easy and interactive way to monitor business performance. Users can analyze sales trends, compare regions, evaluate product performance, and make informed decisions using the dashboard's visual reports.